Part C

In this part of the test, there are two texts about different aspects of healthcare. For **questions 7-22**, choose the answer (**A**, **B**, **C** or **D**) which you think fits best according to the text.

Text 1: Phobia pills

An irrational fear, or phobia, can cause the heart to pound and the pulse to race. It can lead to a full-blown panic attack – and yet the sufferer is not in any real peril. All it takes is a glimpse of, for example, a spider's web for the mind and body to race into panicked overdrive. These fears are difficult to conquer, largely because, although there are no treatment guidelines specifically about phobias, the traditional way of helping the sufferer is to expose them to the fear numerous times. Through the cumulative effect of these experiences, sufferers should eventually feel an increasing sense of control over their phobia. For some people, the process is too protracted, but there may be a short cut. Drugs that work to boost learning may help someone with a phobia to 'detrain' their brain, losing the fearful associations that fuel the panic.

The brain's extraordinary ability to store new memories and forge associations is so well celebrated that its **dark side** is often disregarded. A feeling of contentment is easily evoked when we see a photo of loved ones, though the memory may sometimes be more idealised than exact. In the case of a phobia, however, a nasty experience with, say, spiders, that once triggered a panicked reaction, leads the feelings to resurge whenever the relevant cue is seen again. The current approach is exposure therapy, which uses a process called extinction learning. This involves people being gradually exposed to whatever triggers their phobia until they feel at ease with it. As the individual becomes more comfortable with each situation, the brain automatically creates a new memory – one that links the cue with reduced feelings of anxiety, rather than the sensations that mark the onset of a panic attack.

Unfortunately, while it is relatively easy to create a fear-based memory, expunging that fear is more complicated. Each exposure trial will involve a certain degree of distress in the patient, and although the process is carefully managed throughout to limit this, some psychotherapists have concluded that the treatment is unethical. Neuroscientists have been looking for new ways to speed up extinction learning **for that same reason**.

One such avenue is the use of 'cognitive enhancers' such as a drug called D-cycloserine or DCS. DCS slots into part of the brain's 'NMDA receptor' and seems to modulate the neurons' ability to adjust their signalling in response to events. This tuning of a neuron's firing is thought to be one of the key ways the brain stores memories, and, at very low doses, DCS appears to boost that process, improving our ability to learn. In 2004, a team from Emory University in Atlanta, USA, tested whether DCS could also help people with phobias. A pilot trial was conducted on 28 people undergoing specific exposure therapy for acrophobia – a fear of heights. Results showed that those given a small amount of DCS alongside their regular therapy were able to reduce their phobia to a greater extent than those given a placebo. Since then, other groups have replicated the finding in further trials.

For people undergoing exposure therapy, achieving just one of the steps on the long journey to overcoming their fears requires considerable perseverance, says Cristian Sirbu, a behavioural scientist and psychologist. Thanks to improvement being so slow, patients – often already anxious – tend to feel they have failed. But Sirbu thinks that DCS may make it possible to tackle the problem in a single 3-hour session, which is enough for the patient to make real headway and to leave with a feeling of satisfaction. However, some people have misgivings about this approach, claiming that as it doesn't directly undo the fearful response which is deep-seated in the memory, there is a very real risk of relapse.

Rather than simply attempting to overlay the fearful associations with new ones, Merel Kindt at the University of Amsterdam is instead trying to alter the associations at source. Kindt's studies into anxiety disorders are based on the idea that memories are not only vulnerable to alteration when they're first laid down, but, of key importance, also at later retrieval. This allows for memories to be 'updated', and these amended memories are re-consolidated by the effect of proteins which alter synaptic responses, thereby maintaining the strength of feeling associated with the original memory. Kindt's team has produced encouraging results with arachnophobic patients by giving them propranolol, a well-known and well-tolerated beta-blocker drug, while they looked at spiders. This blocked the effects of norepinephrine in the brain, disrupting the way the memory was put back into storage after being retrieved, as part of the process of reconsolidation. Participants reported that while they still don't like spiders, they were able to approach them. Kindt reports that the benefit was still there three months after the test ended.

Text 1: Questions 7-14

7. In the first paragraph, the writer says that conventional management of phobias can be problematic because of

(A) the lasting psychological effects of the treatment.

(B) the time required to identify the cause of the phobia.

(C) the limited choice of therapies available to professionals.

(D) the need for the phobia to be confronted repeatedly over time.

8. In the second paragraph, the writer uses the phrase '**dark side**' to reinforce the idea that

(A) memories of agreeable events tend to be inaccurate.

(B) positive memories can be negatively distorted over time.

(C) unhappy memories are often more detailed than happy ones.

(D) unpleasant memories are aroused in response to certain prompts.

9. In the second paragraph, extinction learning is explained as a process which

(A) makes use of an innate function of the brain.

(B) encourages patients to analyse their particular fears.

(C) shows patients how to react when having a panic attack.

(D) focuses on a previously little-understood part of the brain.

10. What does the phrase '**for that same reason**' refer to?

(A) the anxiety that patients feel during therapy

(B) complaints from patients who feel unsupported

(C) the conflicting ethical concerns of neuroscientists

(D) psychotherapists who take on unsuitable patients

11. In the fourth paragraph, we learn that the drug called DCS

(A) is unsafe to use except in small quantities.

(B) helps to control only certain types of phobias.

(C) affects how neurons in the brain react to stimuli.

(D) increases the emotional impact of certain events.

12. In the fifth paragraph, some critics believe that one drawback of using DCS is that

(A) its benefits are likely to be of limited duration.

(B) it is only helpful for certain types of personality.

(C) few patients are likely to complete the course of treatment.

(D) patients feel discouraged by their apparent lack of progress.

13. In the final paragraph, we learn that Kindt's studies into anxiety disorders focused on how

(A) proteins can affect memory retrieval.

(B) memories are superimposed on each other.

(C) negative memories can be reduced in frequency.

(D) the emotional force of a memory is naturally retained.

14. The writer suggests that propranolol may

(A) not offer a permanent solution for patients' phobias.

(B) increase patients' tolerance of key triggers.

(C) produce some beneficial side-effects.

(D) be inappropriate for certain phobias.